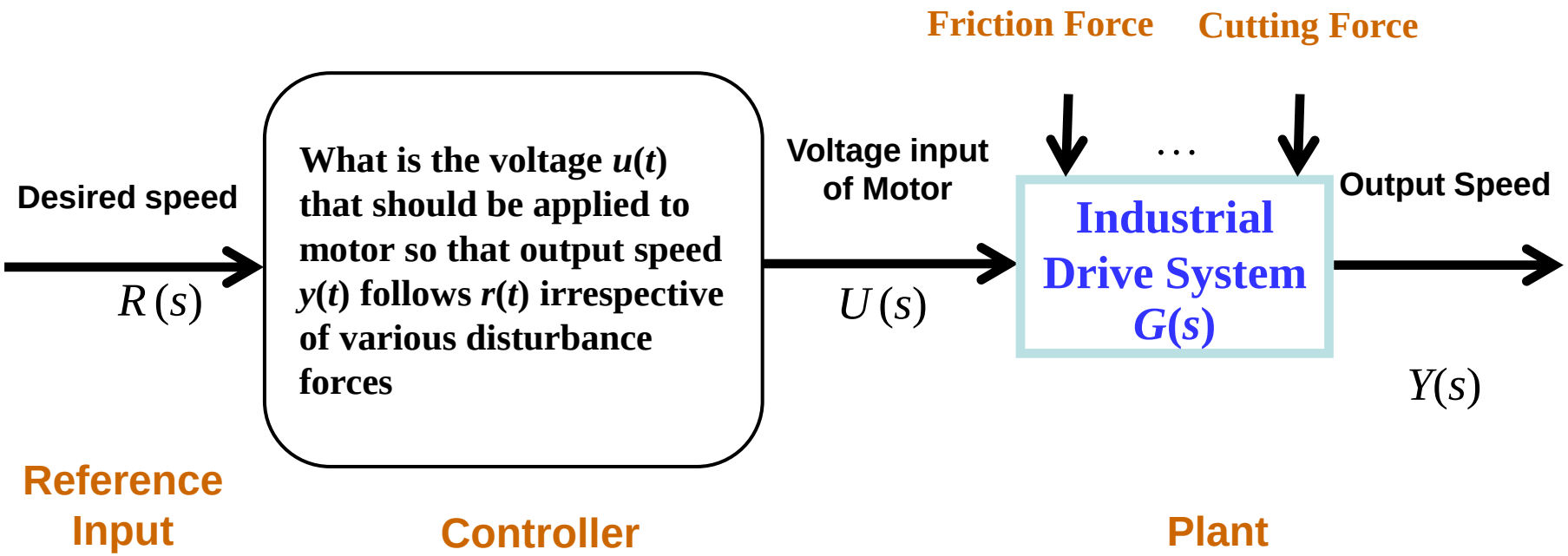
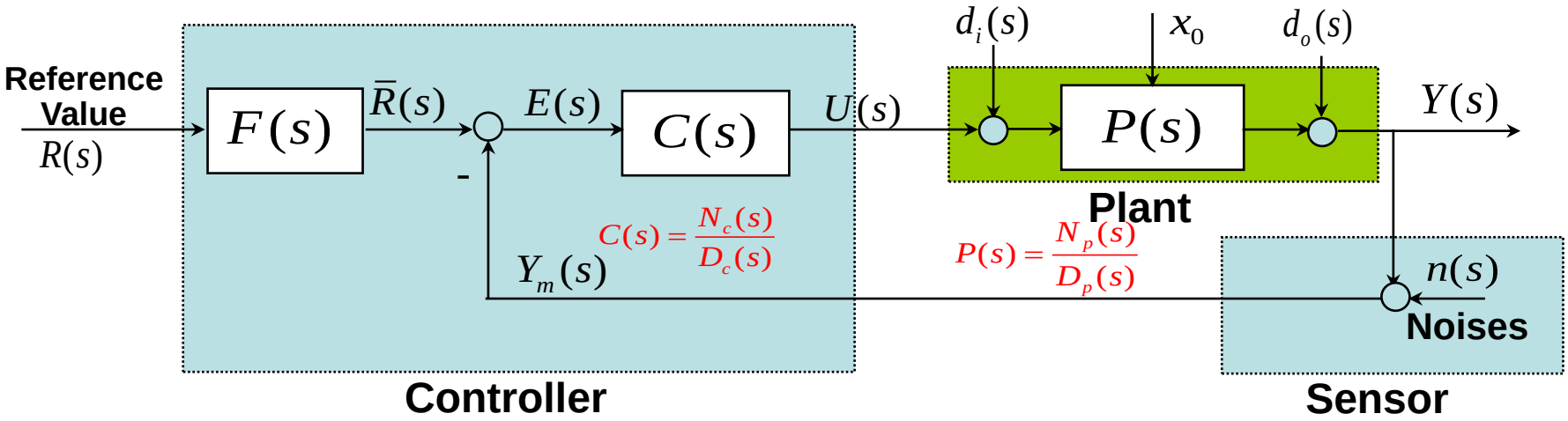
Feedback Control of Mechatronic Systems

- Objectives of Control System Design
- Performance Specifications
- Feedback Control Design Process
- Root Locus Design
- Pole Placement Design

Object of Control System Design



Two-DOF Closed-Loop System



$$U(s)=C(s)E(s)=C(s)[\bar{R}(s)-Y_m(s)]=C(s)\{F(s)R(s)-[Y(s)+n(s)]\}$$

$$Y(s)=d_0+P(s)[U(s)+d_i]+\frac{f(s,x_0)}{D_p(s)}$$

$$=d_0+P(s)C(s)F(s)R(s)-P(s)C(s)Y(s)-P(s)C(s)n(s)+P(s)d_i+\frac{f(s,x_0)}{D_p(s)}$$

$$\Rightarrow\quad [1+P(s)C(s)]Y(s)=d_0(s)+P(s)C(s)F(s)R(s)-P(s)C(s)n(s)+P(s)d_i(s)+\frac{f(s,x_0)}{D_p(s)}$$

Closed-Loop Transfer Functions

$$Y(s)=\underbrace{\frac{P(s)C(s)F(s)}{1+P(s)C(s)}}_{G_{YR}(s)}R(s)+\underbrace{\frac{1}{1+P(s)C(s)}}_{G_{Yd0}(s)}[d_0+\frac{f(s,x_0)}{D_p(s)}]$$

$$+\underbrace{\frac{P(s)}{1+P(s)C(s)}}_{G_{Ydi}(s)}d_i(s)+\underbrace{\frac{-P(s)C(s)}{1+P(s)C(s)}}_{G_{Yn}(s)}n(s)$$

$$U(s)=\frac{C(s)F(s)}{1+P(s)C(s)}R(s)+\frac{-C(s)}{1+P(s)C(s)}[d_0+\frac{f(s,x_0)}{D_p(s)}]$$

$$+\frac{-P(s)C(s)}{1+P(s)C(s)}d_i(s)+\frac{-C(s)}{1+P(s)C(s)}n(s)$$

Sensitivity Functions

$$Y(s)=T(s)F(s)R(s)+S(s)[d_0+\frac{f(s,x_0)}{D_p(s)}]+S_i(s)d_i(s)-T(s)n(s)$$

$$U(s)=S_u(s)[F(s)R(s)-d_0(s)-\frac{f(s,x_0)}{D_p(s)}-P(s)d_i(s)-n(s)]$$

$$T(s)=\frac{P(s)C(s)}{1+P(s)C(s)}=\frac{N_p(s)N_c(s)}{D_p(s)D_c(s)+N_p(s)N_c(s)}:$$ **Complementary sensitivity function**

$$S(s)=\frac{1}{1+P(s)C(s)}=\frac{D_p(s)D_c(s)}{D_p(s)D_c(s)+N_p(s)N_c(s)}:$$ **Sensitivity function**

$$S_i(s)=\frac{P(s)}{1+P(s)C(s)}=\frac{N_p(s)D_c(s)}{D_p(s)D_c(s)+N_p(s)N_c(s)}:$$ **Input disturbance sensitivity function**

$$S_u(s)=\frac{C(s)}{1+P(s)C(s)}=\frac{D_p(s)N_c(s)}{D_p(s)D_c(s)+N_p(s)N_c(s)}:$$ **Control sensitivity function**

Design Trade-off of Sensitivity Functions

- **Relationship Between Sensitivities**

$$S(s)+T(s)=1$$

$$S_i(s)=S(s)P(s)$$

$$S_u(s)=S(s)C(s)$$

- **Practical Implications**

To minimize the effect of noise:

Want : $T(s)=0$

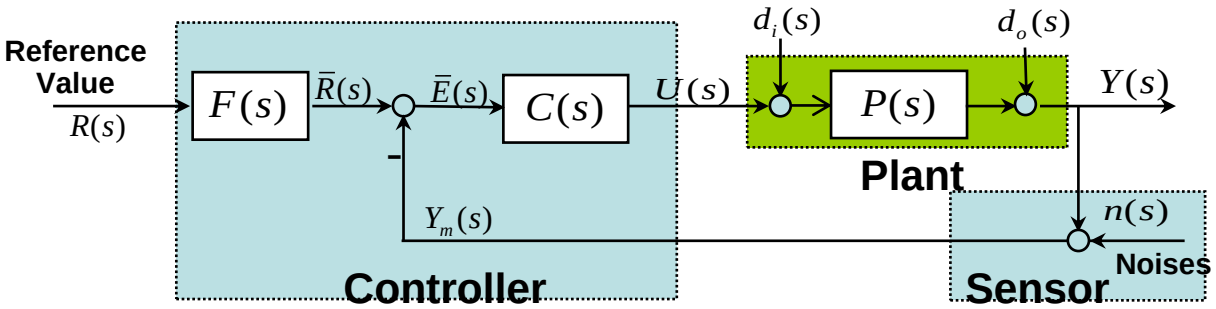
To minimize disturbance effect:

Want : $S(s)=0$

To follow reference:

Want : $T(s)F(s)=1$

Tracking Error of Two-DOF Controller



$L(s) = P(s)C(s)$, open-loop TF

- **CL Response in s-Domain**

$$Y(s) = \underbrace{T(s)F(s)}_{T_r(s)} R(s) + S_i(s)d_i(s) + S(s)d_o(s) - T(s)n(s)$$

$$S(s) = \frac{1}{1+L(s)} = \frac{D_p(s)D_c(s)}{D_p(s)D_c(s)+N_p(s)N_c(s)}$$

$$T(s) = \frac{L(s)}{1+L(s)} = \frac{N_p(s)N_c(s)}{D_p(s)D_c(s)+N_p(s)N_c(s)}$$

$$S_i(s) = \frac{P(s)}{1+L(s)} = \frac{N_p(s)D_c(s)}{D_p(s)D_c(s)+N_p(s)N_c(s)}$$

- **Tracking Error in s-Domain**

$$E(s) = R(s) - Y(s)$$

$$= \underbrace{(1-T_r(s))R(s)}_{E_r(s)} \quad \underbrace{-S_i(s)d_i(s)}_{E_{di}(s)} \quad \underbrace{-S(s)d_o(s)}_{E_{do}(s)} \quad \underbrace{+T(s)n(s)}_{E_n(s)}$$

error due to reference input | error due to input disturbance | error due to output disturbance | error due to measurement noise

Steady-State Error in Time Domain

- **Constant Steady-State Error**

  By assuming that conditions of the Final Value Theorem (FVT) are satisfied, i.e., all poles of $sE(s)$ are in LHP, then, the tracking error $e(t)$ converges to the constant steady-state value e_{ss} given by

$$e_{ss} = \lim_{s \to 0} sE(s)$$

- **Steady-State Error for Sinusoidal Inputs**

$$r(t) = A_r \sin(\omega_r t), \quad d_i(t) = A_{di} \sin(\omega_d t),$$

$$d_o(t) = A_{do} \sin(\omega_d t), \quad n(t) = A_n \sin(\omega_n t)$$

$$\Rightarrow \quad e_{ss}(t) = \underbrace{\left|1 - T_r(j\omega_r)\right| A_r \sin\left(\omega_r t + \angle\left(1 - T_r(j\omega_r)\right)\right)}_{e_{rss}(t)} \underbrace{-\left|S_i(j\omega_d)\right| A_{di} \sin\left(\omega_d t + \angle S_i(j\omega_d)\right)}_{e_{diss}(t)}$$

$$\underbrace{-\left|S(j\omega_d)\right| A_{do} \sin\left(\omega_d t + \angle S(j\omega_d)\right)}_{e_{doss}(t)} \underbrace{+\left|T(j\omega_n)\right| A_n \sin\left(\omega_n t + \angle T(j\omega_n)\right)}_{e_{nss}(t)}$$

Performance Specifications

With a two DOF controller, the closed-loop system response is:

$$Y(s) = T_r(s)R(s) + S_i(s)d_i(s) + S(s)d_o(s) - T(s)n(s)$$

The objective of control system design is to choose the controller TF $C(s)$ and $F(s)$ such that certain performance (design) specifications are met. We want:

(i) The output $y(t)$ to follow the reference input $r(t)$

This is equivalent to specifying that

Ideally, want

$$T_r(s) \equiv 1$$

In reality, try to achieve

$$T_r(j\omega) \approx 1, \quad \omega \in \underbrace{\left[\omega_{low}, \quad \omega_{high}\right]}_{\text{Frequency Range of Ref. Input}}$$

Performance Specifications

With a two DOF controller, the closed-loop system response is:

$$Y(s) = T_r(s)R(s) + S_i(s)d_i(s) + S(s)d_o(s) - T(s)n(s)$$

(ii) The disturbances to have very little effect on the output $y(t)$ within the frequency range where disturbances are most likely to occur

This is equivalent to specifying that

Ideally, want

$$S(s) \equiv 0$$

In reality, try to achieve

$$S(j\omega) \approx 0, \quad \omega \in \underbrace{\left[\omega_{d_low}, \quad \omega_{d_high}\right]}_{\text{Frequency Range of Disturbances}}$$

(iii) The noise to have very little effect on the output $y(t)$ within the frequency range where noises are most likely to occur

This is equivalent to specifying that

Ideally, want

$$T(s) \equiv 0$$

In reality, try to achieve

$$T(j\omega) \approx 0, \quad \omega \in \underbrace{\left[\omega_{n_low}, \quad \infty\right]}_{\text{Frequency Range of Noises}}$$

Steady-State Performance Specifications

- **When Dealing with Polynomial Type Inputs in Time-Domain**

Specific requirements on the system types and the corresponding error constants

➢For constant reference input $r(t)=r_{ss}$, the steady state response $y_{SS}(t)$ to be within an $X\%$ band of the reference input $r(t)$, i.e :

$$\left|\frac{r(t)-y_{SS}(t)}{r(t)}\right| \le X\% \quad \Rightarrow \quad \left|\frac{r_{ss}-T_r(0)r_{ss}}{r_{ss}}\right| < X\% \quad \Leftrightarrow \quad 1-X\% < T_r(0) < 1+X\%$$

or equivalently, the system type w.r.t. reference input should be at least type 0 with the position error constant no less than some value :

$$\left|\frac{1}{1+K_p}\right| < X\% \quad \Leftrightarrow \quad K_p > \frac{1}{X\%}-1$$

➢For unit ramp reference input $r(t)=t$: the system type w.r.t. reference input should be at least type 1 with the position error constant no less than some value:

$$\left|\frac{1}{K_v}\right| < X\% \quad \Leftrightarrow \quad K_v > \frac{1}{X\%}$$

Steady-State Performance Specifications

- **When Dealing with Sinusoidal Type Inputs in Frequency-Domain**

Specific requirements on the CL frequency responses

- The steady-state error $e_{SS}(t)$ to track a reference input of $r(t) = A_r \sin(\omega_r t)$ should be within $X\%$:

$$e_{rss}(t) = \left|1 - T_r(j\omega_r)\right| A_r \sin\left(\omega_r t + \angle\left(1 - T_r(j\omega_r)\right)\right) \quad \Rightarrow \quad \left|1 - T_r(j\omega_r)\right| < X\%$$

- The input disturbance of $d_i(t) = A_{di} \sin(\omega_d t)$ should be attenuated by a factor of γ:

$$e_{diss}(t) = -\left|S_i(j\omega_d)\right| A_{di} \sin\left(\omega_d t + \angle S_i(j\omega_d)\right) \quad \Rightarrow \quad \left|S_i(j\omega_d)\right| < \frac{1}{\gamma}$$

- The output disturbance of $d_o(t) = A_{do} \sin(\omega_d t)$ should be attenuated by a factor of γ:

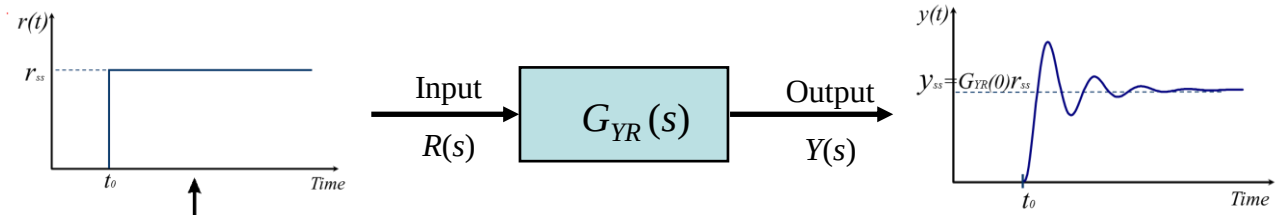
$$e_{doss}(t) = -\left|S(j\omega_d)\right| A_{do} \sin\left(\omega_d t + \angle S(j\omega_d)\right) \quad \Rightarrow \quad \left|S(j\omega_d)\right| < \frac{1}{\gamma}$$

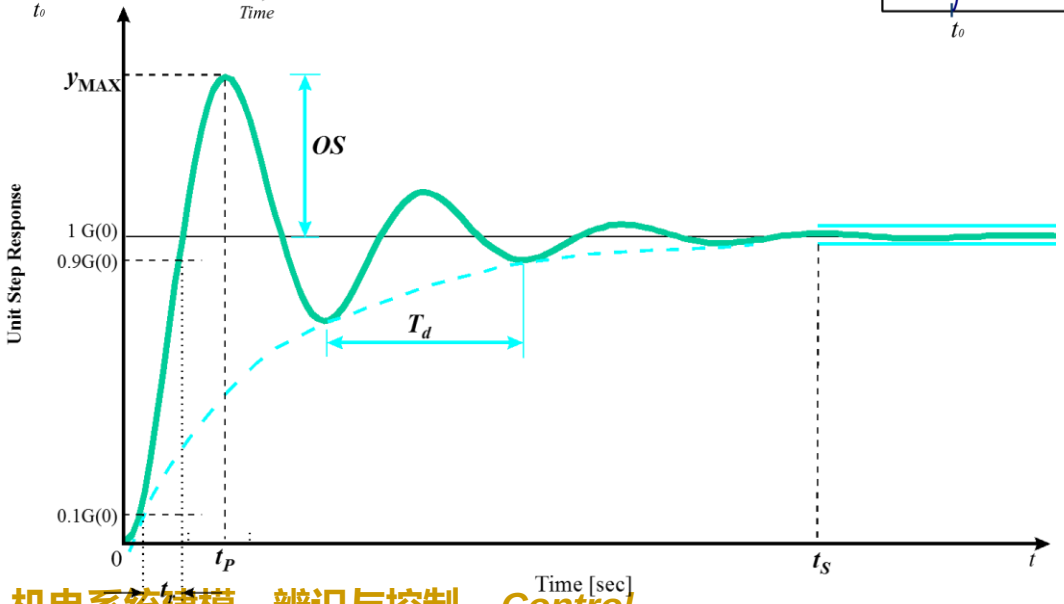
- The noise of $n(t) = A_n \sin(\omega_n t)$ should be attenuated by a factor of γ:

$$e_{nss}(t) = -\left|T(j\omega_n)\right| A_n \sin\left(\omega_n t + \angle T(j\omega_n)\right) \quad \Rightarrow \quad \left|T(j\omega_n)\right| < \frac{1}{\gamma}$$

Transient Performance Specifications

- **Traditionally Specified Based on Unit Step Response**





Transient Performance Specifications

- **Transient Performance (Transient Response)**

Transient performance is usually specified using the unit step response of the system

– **(i) Settling Time (t_S)**

Specifies the time required for the response to reach and stay within a specific percent of the final (steady-state) value

For a 2nd order system without zero

$$p_{1,2} = \underbrace{-\zeta\omega_n}_{\sigma} \pm j\underbrace{\omega_n\sqrt{(\zeta^2-1)}}_{\omega_d} \Longrightarrow \begin{cases} \dfrac{4}{\zeta\omega_n} & \text{for 2\% bound} \\ \dfrac{5}{\zeta\omega_n} & \text{for 1\% bound} \end{cases} \leq \text{Desired Settling Time } (T_{sd})$$

– **(ii)% Overshoot** (%**OS**) (2nd order system without zero)

$$\%OS = 100\% \times e^{-\pi\frac{\zeta}{\sqrt{1-\zeta^2}}} \leq X\%$$

Transient Performance Specifications and CLTF Poles

Assuming that the unit step response of a feedback control system can be approximated by a second-order system without zeros, the requirement on the transient response of the feedback control system can be met by constraining the locations of the poles of the closed-loop transfer function (CLTF) at certain regions on the complex plane :

Dominant CL Poles

$$s_{1,2} = \underbrace{-\zeta\omega_n}_{\sigma} \pm j\underbrace{\omega_n\sqrt{1-\zeta^2}}_{\omega_d} = -\sigma \pm j\omega_d \cdot$$

Settling Time (2%) → ***Puts constraint on the real part of the dominating closed-loop poles***

$$t_S(2\%) = \frac{4}{\zeta\omega_n} = \frac{4}{\sigma} \le T_{sd} \quad \Rightarrow \quad \sigma \ge \frac{4}{T_{sd}}$$

%OS → ***Puts constraint on the damping ratio ζ or the angle φ of the dominating closed-loop poles***

$$\%OS = 100\% \times e^{-\frac{\pi\zeta}{\sqrt{1-\zeta^2}}} \le X\% \quad \Rightarrow \quad \zeta \ge \sqrt{\frac{\left[\ln\left(\frac{X}{100}\right)\right]^2}{\pi^2 + \left[\ln\left(\frac{X}{100}\right)\right]^2}}$$

Performance Specification → CL Pole Locations

- **Transient Performance Specifications and CLTF Pole Locations**

Transient performance specifications can be interpreted as constraints on the locations of the poles of the closed-loop transfer function. Let a pair of closed-loop poles be represented as:

$$p_{1,2} = -\sigma \pm j\omega_d$$

Transient Performance Specifications :

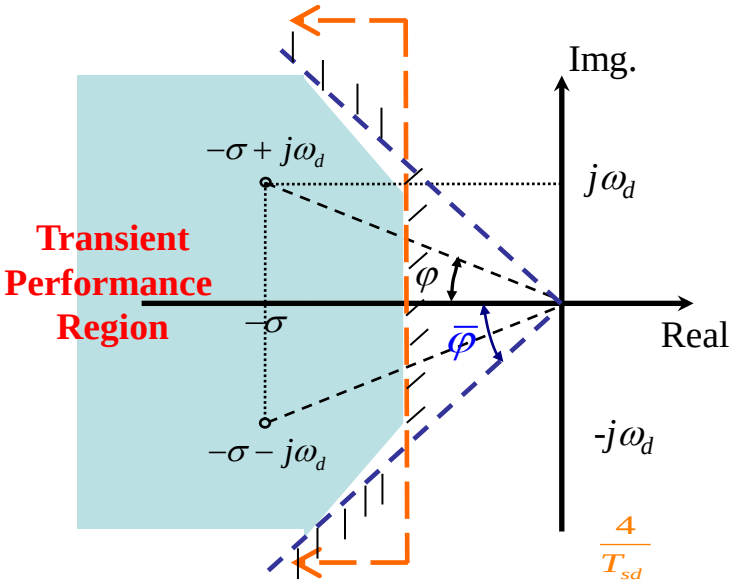
— *Settling Time (2 %)* $\leq T_{sd}$

$$t_S(2\%) = \frac{4}{\sigma} \leq T_{sd} \quad \Rightarrow \quad \sigma \geq \frac{4}{T_{sd}}$$

— $\%OS \leq X\%$

$$\%\mathrm{OS} = 100\% \times e^{-\pi \cdot \frac{\sigma}{\omega_d}} \leq X\%$$

$$\Rightarrow \tan(\varphi) \leq -\frac{\pi}{\ln(X\%)} \Leftrightarrow \varphi \leq \underbrace{\arctan\left(-\frac{\pi}{\ln(X\%)}\right)}_{\bar{\varphi}}$$



Example

A DC motor driven positioning system can be modeled by a second order transfer function:

$$G(s) = \frac{3}{s(s+6)}$$

A proportional feedback control is proposed. It is desired that:

- *(P1) for a unit step response, the steady state position should be within 2% of the desired position,*
- *(P2) the 2% settling time should be less than 2 sec, and*
- *(P3) the percent overshoot should be less than 10%.*

Find

(1) the condition on the proportional gain such that the steady state performance is satisfied;

(2) the allowable region in the complex plane for the closed-loop poles.

Find closed-loop transfer function:

$$G_{YR}(s) = \frac{C(s)G(s)}{1+C(s)G(s)} = \frac{3K}{s(s+6)+3K}$$

$$\Rightarrow p_{1,2} = -3 \pm \sqrt{9-3K}$$

Write down the performance specifications:

To meet,

$$(P1): \quad \left|\frac{\theta_{ss}-\theta_d}{\theta_d}\right| \le 2\% = 0.02, \quad \left|\frac{G_{YR}(0)\theta_d-\theta_d}{\theta_d}\right| \le 0.02$$

$$\Rightarrow \quad 0.98 \le G_{YR}(0) \le 1.2$$

$$(P2): \quad t_s \le 2\,\text{sec} \Leftrightarrow \sigma \ge \frac{4}{t_s} = 2$$

$$(P3): \quad OS\% \le 10\% \Rightarrow 100e^{-\frac{\pi\zeta}{\sqrt{1-\zeta^2}}} \le 10\%$$

$$\Rightarrow \quad \zeta \ge \sqrt{\frac{\left[\ln\left(\frac{X}{100}\right)\right]^2}{\pi^2+\left[\ln\left(\frac{X}{100}\right)\right]^2}} = 0.5912$$

Example

Steady state performance constraint:

$$0.98 \le G_{YR}(0) = 1 \le 1.02$$

Transient performance constraint:

2% Settling Time

$$\sigma \ge 2$$

Percent Overshoot (%OS)

$$OS\% \le 10\%$$

$$\varphi \le \bar{\varphi} = \arctan\left(-\frac{\pi}{\ln(X\%)}\right)$$

$$= \tan^{-1}\left(-\frac{\pi}{\ln(0.1)}\right) = 53.76^{o}$$

Closed-Loop Poles ($K>3$):

$$s^2 + 6s + 3K = 0 \quad \Rightarrow \quad s_{1,2} = \underbrace{-3}_{\sigma} \pm j\underbrace{\sqrt{3K-9}}_{\omega_d}$$

$$\therefore \quad \frac{\sqrt{3K-9}}{3} = \tan(\varphi) < \tan(\bar{\varphi}) \quad \Rightarrow \quad K < 8.6$$

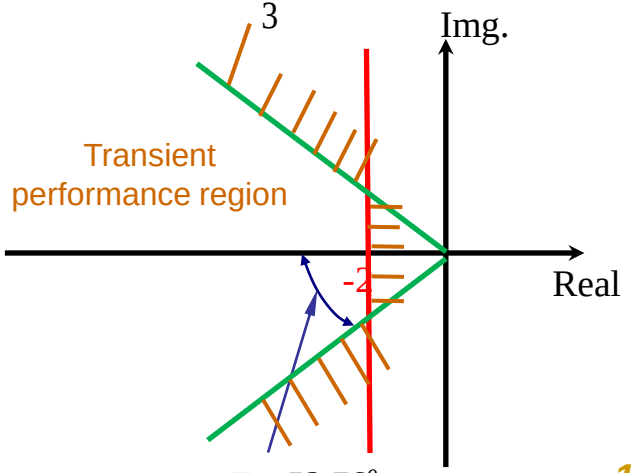
Closed-Loop Poles ($K<3$):

$$s^2 + 6s + 3K = 0 \quad \Rightarrow \quad s_{1,2} = -3 \pm \sqrt{9-3K}$$

$$\therefore \quad -3 + \sqrt{9-3K} < -2 \quad \Rightarrow \quad K > \frac{8}{3}$$

For CL Poles to be in Transient Performance Region :

$$\Rightarrow \frac{8}{3} < K < 8.6$$



$$\bar{\varphi} = 53.76^{o}$$

Feedback Control Design Process

(1)Model the physical system (plant) that we want to control and obtain its I/O transfer function $P(s)$

(2)Determine sensor dynamics and actuator dynamics (if necessary)

(3)Draw the closed-loop block diagram, which includes the plant, sensor, actuator and controller transfer functions $C(s)$ and $F(s)$

(4)Obtain the closed-loop transfer functions

(5)Based on the performance specifications, find the conditions that the CLTFs have to satisfy

(6)Choose controller structures for $C(s)$ and $F(s)$ and substitute them into the CLTFs

(7)Select the controller parameters so that the design constraints are satisfied

(8)Verify your design via computer simulation and actual implementation

Main Contents of Course

Mechanical | Electrical | Electro-Mechanical | Fluid | Thermal

Dynamic Modeling

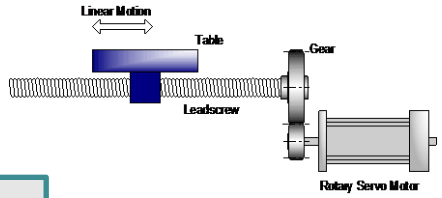
Structural selection

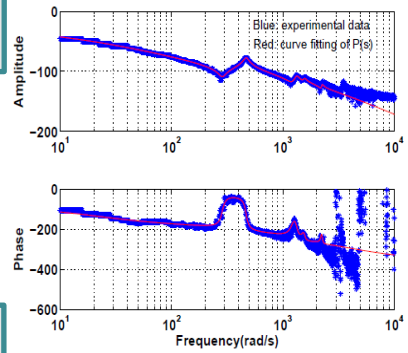
System Identification

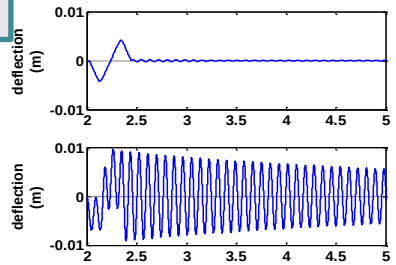
Parameter estimation

Control System Design

Root-locus design | Pole placement design | Frequency design | Architectural issues | Design limitations | State-feedback



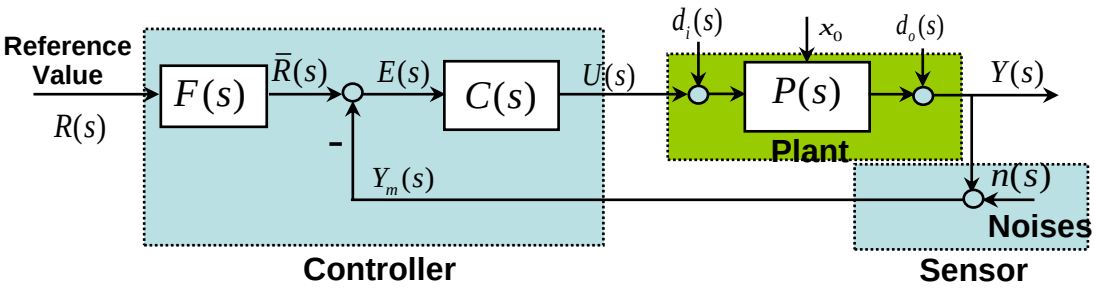




Principle of Controller Design via Root Locus

- **Underlying Design Principles**

  Choosing a feedback controller TF such that all the dominant CL poles are in the performance region to meet the transient performance specifications:



$$P(s)=\frac{N_p(s)}{D_p(s)}=\frac{b_{pm_p}\left(s-z_{p1}\right)\cdots\left(s-z_{pm_p}\right)}{\left(s-p_{p1}\right)\cdots\left(s-p_{pn_p}\right)}$$

$$C(s)=K_c\frac{N_c(s)}{D_c(s)}=K_c\frac{\left(s-z_{c1}\right)\cdots\left(s-z_{cm_c}\right)}{\left(s-p_{c1}\right)\cdots\left(s-p_{cn_c}\right)}$$

Closed-Loop Poles :

$$\underbrace{D_p(s)D_c(s)}_{D(s)}+\underbrace{K_c}_{K}\underbrace{N_p(s)N_c(s)}_{N(s)}=0 \qquad \text{or} \qquad 1+K\frac{N(s)}{D(s)}=0\Leftrightarrow 1+L(s)=0$$

Open-loop poles :

$$D(s)=0 \quad\Leftrightarrow\quad \underbrace{D_p(s)}_{\text{plant TF poles}}=0 \quad or \quad \underbrace{D_c(s)}_{\text{controller TF poles}}=0$$

Open-loop zeros :

$$N(s)=0 \quad\Leftrightarrow\quad \underbrace{N_p(s)}_{\text{plant zeros}} \quad or \quad \underbrace{N_c(s)}_{\text{controller TF zeros}}=0$$

Principle of Controller Design via Root Locus

⇒ **Role of Zeros of the Controller TF $C(s)$ (i.e., Roots of $N_c(s)$):**

Roots of $N_c(s)$ are also roots of $N(s)$ ⇒ add open-loop zeros

Role of Poles of the Controller TF $C(s)$ (i.e., Roots of $D_c(s)$) :

Roots of $D_c(s)$ are also roots of $D(s)$ ⇒ add open-loop poles

- **Design Procedure**

Step1: Pick up $N_c(s)$ and $D_c(s)$ (i.e., choosing the additional open-loop zeros and poles that are needed) such that the resulting Root Locus intersects performance region or pass through the desired CL pole locations; Angle Criterion may be used in determining the locations of the needed additional open-loop poles and zeros.

Step2: Pick up a gain K_c such that all CL poles are in the performance region or at the desired locations; Magnitude Criterion should be used to find the gain K_c corresponding to the desired CL poles.

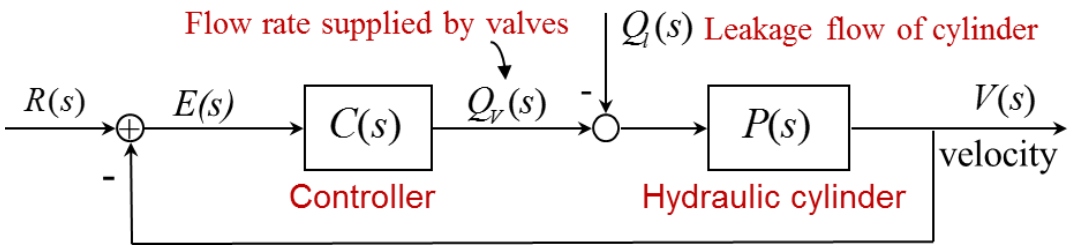
Ex.RL1 Motion Control of Hydraulic Cylinders

Hydraulic excavators exhibit severe chattering problem since the hydraulic cylinder dynamics is described by a lightly damped second-order system. You are required to design a feedback controller to solve this problem by satisfying the following transient performance requirements:

(P1). The resulting CL system should have a OS% less than 4.3%

(P2). 2% settling time t_s should be smaller than 1 second

Block Diagram of Feedback Control System :



$$P(s) = \frac{K_p\omega_p^2}{s^2 + 2\zeta_p\omega_p s + \omega_p^2} = K_p\omega_p^2 \frac{N_p(s)}{D_p(s)}$$

ω_p=3Hz=6π rad/s,

ζ_p=0.1, K_p=0.2

$$\Rightarrow D_p(s) = s^2 + 2\zeta_p\omega_p s + \omega_p^2 = (s-p_1)(s-p_2), \quad p_{1,2} = -1.89 \pm 18.79j$$

Performance Regions :

(P1) OS% ≤ 4.3%:

$$\Rightarrow \varphi = -\tan^{-1}\left(\frac{\pi}{\log(OS\%100)}\right) \text{ or } \zeta \geq \sqrt{\frac{\left[\ln\left(\frac{X}{100}\right)\right]^2}{\pi^2 + \left[\ln\left(\frac{X}{100}\right)\right]^2}}$$

$\Rightarrow \varphi = 45°$

or $\zeta = 0.707$

(P2) $t_s \leq 1$ sec

$\Rightarrow \tau \leq$ ¼ sec $\Rightarrow \sigma \geq 1/\tau = 4$

Ex.RL1 Motion Control of Hydraulic Cylinders

Proportional (P) Feedback Controller :

In s-domain:

$$C(s)=K_c$$

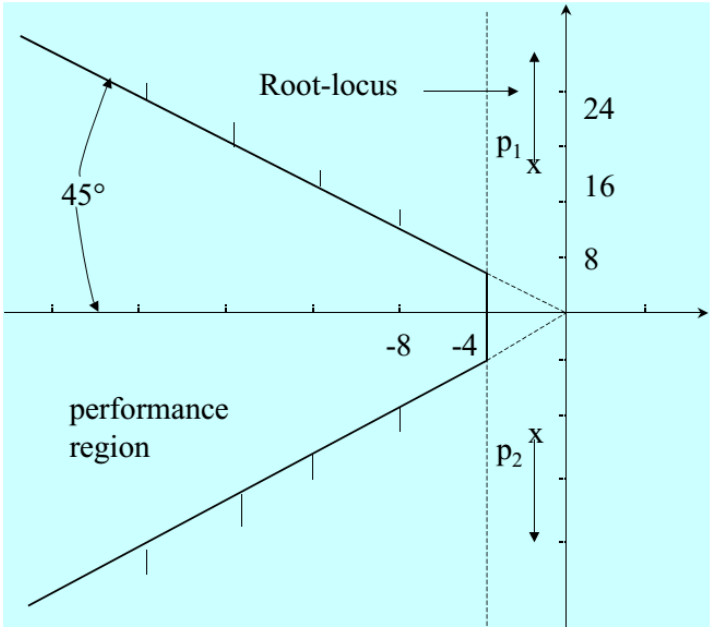
In time-domain:

$$q_V(t) = K_c\, e(t)$$

Root-Locus for P-Control:

$\Rightarrow$

$$\begin{cases} D(s) = D_p(s)D_c(s) = D_p(s) = (s-p_1)(s-p_2) \\ N(s) = N_p(s)N_c(s) = N_p(s) = 1 \\ K = K_c K_p = 71K_c \end{cases}$$



Q: Can we use a proportional (P) feedback controller to achieve our objective?

No. Since the root-locus does not intersect the performance region.

Ex.RL1 Motion Control of Hydraulic Cylinders

Proportional plus Derivative (PD) feedback controller :

Control Law in Time-Domain:

$$q_V(t) = \underbrace{K_{CP}e(t)}_{\substack{\text{proportional}\\ \text{feedback}}} + \underbrace{K_{CD}\dot{e}(t)}_{\substack{\text{derivative}\\ \text{feedback}}} \quad \text{where} \quad \underbrace{e(t) = r(t) - y(t)}_{\text{tracking error}}$$

Control Law in s-Domain:

$$Q_V(s) = K_{CP}E(s) + K_{CD}sE(s) = (K_{CP} + K_{CD}s)\,E(s) = \underbrace{K_{CD}\left(s + \frac{K_{CP}}{K_{CD}}\right)}_{C(s)} E(s)$$

$$C(s) = K_{CD}\left(s + \frac{K_{CP}}{K_{CD}}\right) = K_C(s+\alpha)$$

where $K_C = K_{CD}$, $N_c(s) = s + \alpha$, $D_c(s) = 1$, $\alpha = \dfrac{K_{CP}}{K_{CD}}$

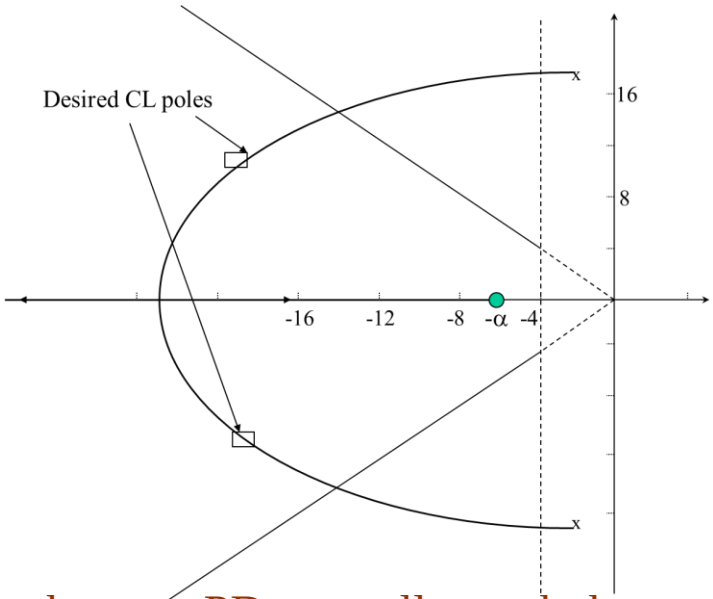
Closed-Loop Characteristic Equation (CLCE):

$D(s) = D_p(s)D_c(s) = D_p(s) = (s - p_1)(s - p_2)$ $\Rightarrow$ two open-loop poles at p_1 and p_2

$N(s) = N_p(s)N_c(s) = (s + \alpha)$ $\Rightarrow$ one open-loop zeros at $-\alpha$

$K = K_cK_p = 71K_c$ CLCE: $D(s) + KN(s) = 0$

Ex.RL1 Motion Control of Hydraulic Cylinders



Q: Can we choose a PD controller such that we can achieve our objective?

Yes. As you can see from the sketch of root-locus, by placing the additional open-loop zero $z_1 = -\alpha$ far away imaginary axis enough, the resulting root-locus will intersect the performance region.

Ex.RL1 Motion Control of Hydraulic Cylinders

Select Controller Gains

For the above PD controller, choose controller gains so that the resulting CL poles are at -18 ± 12*j*, which are in the performance region

Method 1: Direct Pole-Placement

***Step 1**: Obtain the CLCE Expression*

$$D(s)+KN(s)=D_p(s)+K(s+\alpha)=s^2+3.77s+355+K(s+\alpha)$$
$$=s^2+(3.77+K)s+355+K\alpha$$

***Step 2**: Obtain the Desired CLCE from Desired CL Pole Locations*

Want the desired CL poles at $p_{1,2d}$ = -18 ± 12*j*

$\Rightarrow$ Desired CLCE: $(s-p_{1d})(s-p_{2d})=(s+18-12j)(s+18+12j)=s^2+36s+468$

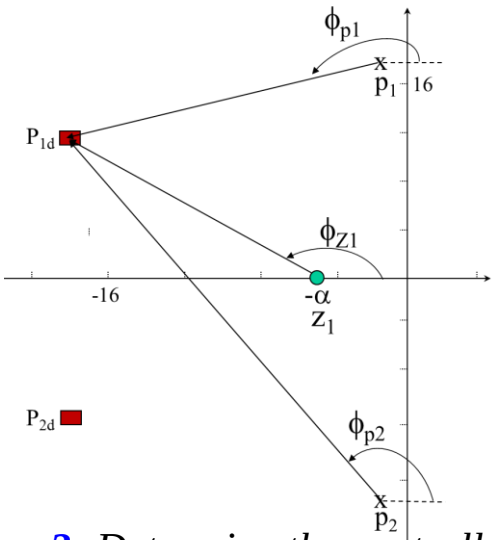
***Step 3**: Match the CLCE expression with the desired CLCE to obtain a set of equations for solving unknown controller gains*

$$\begin{cases}3.77+K=36\\355+K\alpha=468\end{cases}\Rightarrow\begin{cases}K=32.2\\\alpha=3.5\end{cases}\Rightarrow K_{CD}=K_c=\frac{K}{K_p}=\frac{32.2}{71}=0.45, K_{CP}=K_{CD}\alpha=1.59$$

Ex.RL1 Motion Control of Hydraulic Cylinders

Method 2: Pole-Placement via Root Locus

***Step 1**: Pick up compensator $N_c(s)$ and $D_c(s)$, i.e., choose the additional open-loop poles/zeros so that the resulting root locus pass the desired CL pole locations via Angle Criterion*



$$\phi_{p_1} = \angle(p_{1d} - p_1) = -157^\circ \qquad \phi_{p_2} = \angle(p_{1d} - p_2) = 117.7^\circ$$

$$\phi_{z_1} = \angle(-18 + \alpha + 12j) = 180^\circ - \tan^{-1}\left(\frac{12}{18-\alpha}\right) \qquad \text{(Assume } \alpha < 18\text{)}$$

Angle Criterion

$$\angle(p_{1d} - z_1) - \angle(p_{1d} - p_1) - \angle(p_{1d} - p_2) = 180^\circ$$

$$\downarrow \qquad \downarrow \qquad \downarrow$$

$$\phi_{z1} \qquad \phi_{p1} \qquad \phi_{p2}$$

$$\Rightarrow \phi_{z1} = 180^\circ + \phi_{p1} + \phi_{p2} = 140.04^\circ$$

$$\frac{12}{-18+\alpha} = \tan(\phi_{z1} - 180^\circ) = -0.827$$

$$\Rightarrow \alpha = 3.5$$

***Step 2**: Determine the controller gain K_c via Magnitude Condition*

Magnitude Condition

$$K = \frac{|p_{1d} - p_1| . |p_{1d} - p_2|}{|p_{1d} - z_1|} = \frac{7.5 \times 34}{18.8} = 32.2$$

Ex.RL2 Motion Control of Hydraulic Cylinders

Let us reconsider Ex.RL1. In addition to the transient performance requirements (P1) and (P2), the CL system is required to meet the following steady-state (SS) performance requirement:

(P3). Zero steady-state tracking error for any constant reference velocity

Q: Can the previously designed PD controller meet the SS requirement?

$$T_r(s) = \frac{KN(s)}{D(s)+KN(s)} = \frac{32.2\times(s+3.5)}{s^2+36s+468}$$

since

$$\lim_{s\to 0}\frac{1-T_r(s)}{s^0} = 1-T_r(s) = 1-\frac{32.2\times 3.5}{468} = 0.76$$

The resulting CL system is of type 0 w.r.t. reference input and the steady-state tracking error for a unit step reference input is 0.76. Therefore, PD controller can NOT meet the SS requirement.

Ex.RL2 Motion Control of Hydraulic Cylinders

Proportional plus Derivative plus Integral (PID) feedback controller:

Control Law in Time-Domain

$$q_V(t) = \underbrace{K_{CP} e(t)}_{\text{proportional feedback}} + \underbrace{K_{CD} \dot{e}(t)}_{\text{derivative feedback}} + \underbrace{K_{CI} \int e(t)dt}_{\text{integral feedback}}$$

$$T_r(s) = \frac{KN(s)}{D(s)+KN(s)} = \frac{K_c K_p N_p(s) N_c(s)}{D_p(s)D_c(s)+K_p K_c N_p(s) N_c(s)}$$

$$= \frac{K(s-z_1)(s-z_2)}{s(s-p_1)(s-p_2)+K(s-z_1)(s-z_2)}$$

$$1-T_r(s) = \frac{s(s-p_1)(s-p_2)}{s(s-p_1)(s-p_2)+K(s-z_1)(s-z_2)}$$

Control Law in s-Domain

$$Q_V(s) = K_{CP}\, E(s) + K_{CD}\, sE(s) + K_{CI} \cdot \frac{1}{s} E(s) = \underbrace{(K_{CP} + K_{CD}\, s + K_{CI} \cdot \frac{1}{s})}_{C(s)} E(s)$$

$$C(s) = K_{CP} + K_{CD}\, s + \frac{1}{s} K_{CI} = \frac{K_{CD}\, s^2 + K_{CP}\, s + K_{CI}}{s} = \frac{K_c (s-z_{c1})(s-z_{c2})}{s}$$

$$\begin{aligned} K_{CD} &= K_c \\ K_{CI} &= K_{CD}\, z_{c1} z_{c2} \\ K_{CP} &= -K_{CD}(z_{c1}+z_{c2}) \end{aligned} \Leftrightarrow \begin{cases} K_c = K_{CD} \\ N_c(s) = (s-z_{c1})(s-z_{c2}) \\ D_c(s) = s \end{cases}$$

PID controller adds one additional pole at $p_{c1} = 0$ and two additional open-loop zeros at z_{c1} and z_{c2}

Q: Can PID controller meet the steady-state requirement P3?

Yes. Since $\lim_{s\to 0}\{1-T_r(s)\} = \lim_{s\to 0} \frac{s p_1 p_2}{K z_1 z_2} = 0$ ⟹ The CL system is of type 1 w.r.t. reference input

For any constant reference input, SS error is zero

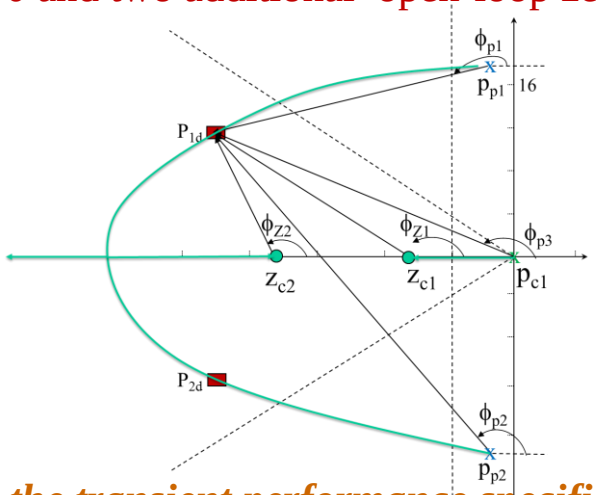
Ex.RL2 Motion Control of Hydraulic Cylinders

Closed-Loop Characteristic Equation(CLCE)

$$D(s)+KN(s)=0 \text{ or } \underbrace{s(s-p_1)(s-p_2)}_{D(s)}+\underbrace{K(s-z_1)(s-z_2)}_{N(s)}=0$$

i.e., PID controller adds one additional pole at p_2= 0 and two additional open-loop zeros at z_1 and z_2

Root-Locus for PID-Control



Q: Can we choose a PID controller to meet the transient performance specification?

Yes. Since the root-locus will intersect performance region as shown in above

Ex.RL2 Motion Control of Hydraulic Cylinders

Select PID Gains:

For the above PID controller, choose controller gains so that the dominant poles of the resulting CL system are at -18 ± 12*j*, which are in the performance region

Method 1: Direct Pole-Placement

***Step 1**: Obtain the CLCE Expression*

$$D(s)+KN(s)=s(s-p_1)(s-p_2)+K(s-z_1)(s-z_2)=s^3+3.77s^2+355s+K[s^2-(z_1+z_2)s+z_1z_2]$$

$$=s^3+(3.77+K)s^2+[355-K(z_1+z_2)]s+Kz_1z_2$$

***Step 2**: Obtain the Desired CLCE from Desired CL Pole Locations*

$p_{1,2d}$ = -18 ± 12*j* , Choose *K*, z_1, z_2 to place the three CL poles at any position as we want (p_{3d}= -20) ⇒ Desired CLCE: $(s-p_{1d})(s-p_{2d})(s-p_{3d})=(s+18-12j)(s+18+12j)(s+20)$

$$=s^3+56s^2+1188s+9360$$

***Step 3**: Match the CLCE expression with the desired CLCE to obtain a set of equations for solving unknown controller gains*

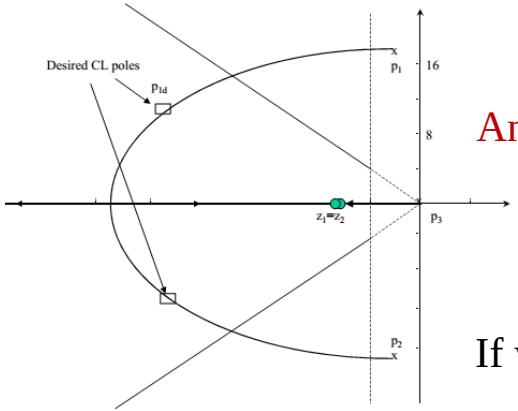
$$\begin{cases} 3.77+K=56 \\ 355-K(z_1+z_2)=1188 \\ Kz_1z_2=9360 \end{cases} \Rightarrow \begin{cases} K=52.23 \\ z_1=-8+10.8j \\ z_2=-8-10.8j \end{cases}$$

$$\Rightarrow K_{CD}=K_c=\frac{K}{K_p}=\frac{52.23}{71}=0.736, K_{CP}=-K_{CD}(z_1+z_2)=11.73, K_{CI}=K_{CD}z_1z_2=131.8$$

Ex.RL2 Motion Control of Hydraulic Cylinders

Method 2: Pole-Placement via Root Locus

***Step 1**: Pick up compensator $N_c(s)$ and $D_c(s)$, i.e., choose the additional open-loop poles/zeros so that the resulting root locus pass the desired CL pole locations via Angle Criterion:*



$$\phi_{p_1} = \angle(p_{1d} - p_1) = -157^\circ \qquad \phi_{p_2} = \angle(p_{1d} - p_2) = 117.7^\circ \qquad \phi_{p_3} = \angle(p_{1d} - p_3) = 146.3^\circ$$

$$\phi_{z_1} = \angle(-18 + z_1 + 12j) = 180^\circ - \tan^{-1}\left(\frac{12}{18-\alpha}\right)$$

Angle Criterion:

$$\angle(p_{1d} - z_1) + \angle(p_{1d} - z_2) - \angle(p_{1d} - p_1) - \angle(p_{1d} - p_2) - \angle(p_{1d} - p_3) = 180^\circ$$

$$\downarrow \qquad \downarrow \qquad \downarrow \qquad \downarrow \qquad \downarrow$$

$$\phi_{z1} \qquad \phi_{z2} \qquad \phi_{p1} \qquad \phi_{p2} \qquad \phi_{p3}$$

$$\Rightarrow \phi_{z1} + \phi_{z2} = 180^\circ + \phi_{p1} + \phi_{p2} + \phi_{p3} = 286.7^\circ$$

Case A: Adding a pair of real zeros

If we choose

$$z_1 = z_2, \text{ then } \phi_{z1} = \phi_{z2} = 143.4^\circ \quad \Rightarrow z_1 = z_2 = -18 - \frac{12}{\tan(\phi_{z1} - 180^\circ)} = -1.86$$

The resulting CL system $\qquad p_{3d} = -0.24,\ p_{1d} = -18 + 12j,\ p_{2d} = -18 - 12j$

may not be able to satisfy setting time requirements

→ not a good design!

Ex.RL2 Motion Control of Hydraulic Cylinders

Method 2: Pole-Placement via Root Locus

***Step 2**: Determine the controller gain K_c via Magnitude Condition*

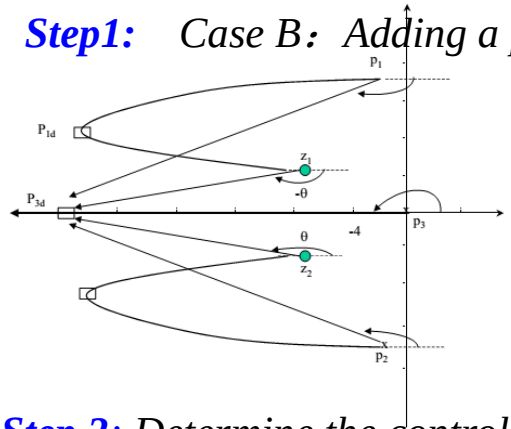
Magnitude Condition for $\boldsymbol{p_{1d}}$:

$$K = \frac{|p_{1d} - p_1| \cdot |p_{1d} - p_2| |p_{1d} - 0|}{|p_{1d} - z_1| |p_{1d} - z_2|} = \frac{17.47 \times 34.7 \times 21.6}{20.1^2} = 32.4$$

Ex.RL2 Motion Control of Hydraulic Cylinders

Method 2: Pole-Placement via Root Locus

Step1: *Case B: Adding a pair of complex zeros*



$$z_1 = -a + bj \text{ and } z_2 = -a - bj$$

Angle Criterion:

$$\angle(p_{3d} - z_1) + \angle(p_{3d} - z_2) - \angle(p_{3d} - p_1) - \angle(p_{3d} - p_2) - \angle(p_{3d} - p_3) = 180^\circ$$

$$\downarrow \qquad \downarrow \qquad \downarrow \qquad \downarrow \qquad \downarrow$$

$$-\theta \qquad \theta \qquad -134^\circ \qquad 134^\circ \qquad 180^\circ$$

Always satisfy Angle Condition!

Step 2: *Determine the controller gain K_c via Magnitude Condition*

Magnitude Condition for $\boldsymbol{p_{1d}}$:

$$K = \frac{|p_{1d} - p_1| \cdot |p_{1d} - p_2||p_{1d} - p_3|}{|p_{1d} - z_1||p_{1d} - z_2|} = \frac{17.47 \times 34.7 \times 21.6}{\sqrt{(-18+a)^2 + (12-b)^2}\sqrt{(-18+a)^2 + (12+b)^2}}$$

Magnitude Condition for $\boldsymbol{p_{3d}}$ **=-20**:

$$K = \frac{|p_{3d} - p_1| \cdot |p_{3d} - p_2||p_{3d} - p_3|}{|p_{3d} - z_1||p_{3d} - z_2|} = \frac{26 \times 26 \times 20}{\left(\sqrt{(-20+a)^2 + b^2}\right)^2}$$

$$\Rightarrow \begin{cases} a = 8 \\ b = 10.8 \\ K = 52.2 \end{cases} \text{ or } \begin{cases} z_{1,2} = -8 \pm 10.8j \\ K = 52.2 \end{cases}$$

Ex.RL2 Motion Control of Hydraulic Cylinders

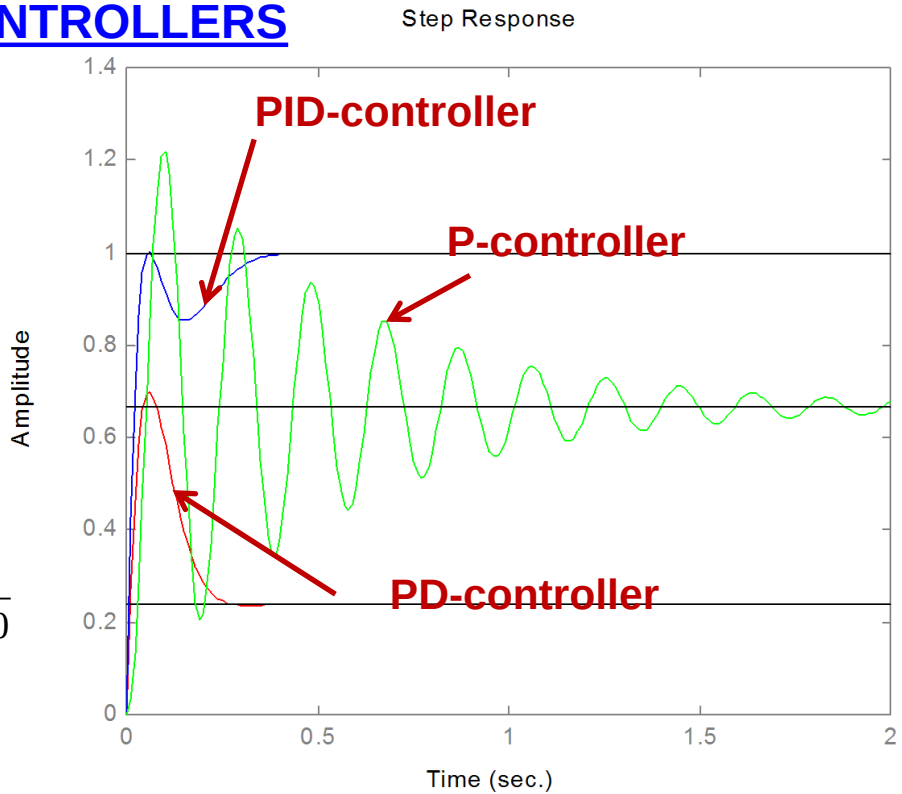
COMPARISON OF P, PD, and PID CONTROLLERS

The closed-loop transfer functions for different controllers are:

P-controller with K_c =10: $$T(s)=\frac{710}{s^2+3.77s+1065}$$

PD-controller: $$T(s)=\frac{32.2(s+3.5)}{s^2+36s+468}$$

PID-controller: $$T(s)=\frac{52.23s^2+833s+9360}{s^3+56s^2+1188s+9360}$$



Unit step responses of the P-controller, PD controller, and PID controller

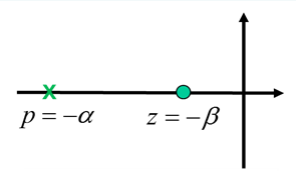
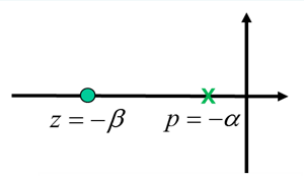
Ex.RL2 Motion Control of Hydraulic Cylinders

SUMMARY OF PID CONTROL

(a) The integral control is to reduce or eliminate constant steady-state errors, but this benefit typically comes at the cost of a degraded transient response if the controller is not well designed.

(b) The derivative feedback is used in conjunction with proportional and/or integral feedback to increase damping and generally improve the stability and the transient performance of a system.

Lead-Lag Compensation Design via Root Locus Principle

	Lead Compensation	Lag Compensation
Controller	$C_{lead}(s) = K_c \dfrac{s+\beta}{s+\alpha}, \quad \beta<\alpha$	$C_{lag}(s) = \dfrac{s+n\alpha_2}{s+\alpha_2} = \dfrac{s+\beta}{s+\alpha}, \quad \beta>\alpha$
Pole and zero location		
Function	Improve the transient performance	Decrease the steady-state error
Prior specifying	*Place the dominant poles according to performance specifications*	*Specify n by meeting the steady-state performance requirement*
Design procedure	*Use Phase Condition, Magnitude Condition or Pole placement method to calculate the control parameters*	*Let the pole-zero pair of the lag compensator close to the origin for not affecting the transient performance*

Main Contents of Course

Mechanical | Electrical | Electro-Mechanical | Fluid | Thermal

Dynamic Modeling

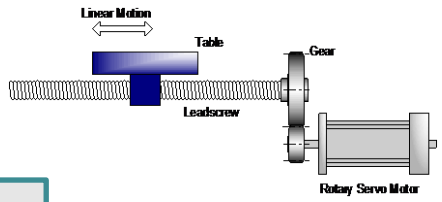
Structural selection

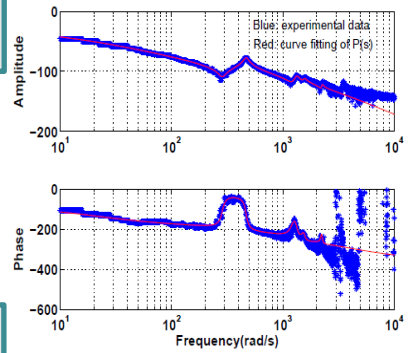
System Identification

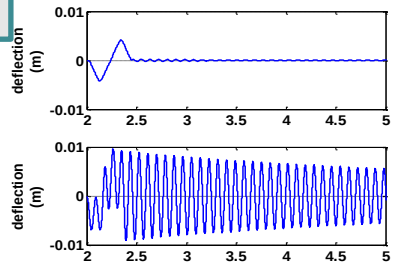
Parameter estimation

Control System Design

Root-locus design | Pole placement design | Frequency design | Architectural issues | Design limitations | State-feedback







Pole Placement

- **Nominal Closed Loop Characteristic Equation**

where $D_{pn}(s)D_c(s)+N_{pn}(s)N_c(s)=0$

$$N_{pn}(s)=b_{pm}s^m+b_{p(m-1)}s^{m-1}+\cdots+b_{p1}s+b_{p0}$$ (assume $m<n$)

$$D_{pn}(s)=s^n+a_{p(n-1)}s^{n-1}+\cdots+a_{p1}s+a_{p0}$$

$$N_c(s)=b_{cm_c}s^{m_c}+b_{c(m_c-1)}s^{m_c-1}+\cdots+b_{c1}s+b_{c0}$$ $m_c<n_c$

$$D_c(s)=s^{n_c}+a_{c(n_c-1)}s^{n_c-1}+\cdots+a_{c1}s+a_{c0}$$

m_c+1 controller coefficients for N_c: $b_{cmc},\ldots,b_{c0}$

n_c controller parameters to choose for D_c: $a_{c(nc-1)},\ldots,a_{c0}$

- **Nominal Closed Loop Characteristic Polynomial**

$$A_C(s)=D_{pn}(s)D_c(s)+N_{pn}(s)N_c(s)$$

$$=s^{n_L}+a_{Cn_L-1}s^{n_L-1}+\cdots+a_{C1}s+a_{C0}$$

Order $m+m_c<n+n_c=n_L$

Pole Placement

- **Desired Closed Loop Characteristic Polynomial**

$$A_{Cd}(s) = (s - p_{1d})(s - p_{2d})\cdots(s - p_{n_L d})$$
$$= s^{n_L} + a_{C(n_L-1)d}\, s^{n_L-1} + \cdots + a_{C1d}\, s + a_{C0d}$$

- **General Formulation of Pole Placement**
  - Given $P_n(s)$ and any desired CL pole locations, i.e., known n_L and $A_{Cd}(s)$ but with coefficients being arbitrarily specified, does there exist a proper $C(s)$ that makes $A_C(s)$ being the desired $A_{Cd}(s)$?

SISO Polynomial Approach

Case 1: $n_L = 2n - 1 + k$

→ The controller order is $n_c = n-1+k, k \geq 0$

Then, we have n_c controller parameters to choose for $D_c(s)$ and m_c+1 parameters for $N_c(s)$

→ A total of $n_c + m_c + 1 = n - 1 + k + m_c + 1$ coefficients free to choose

→(i) $n_c + m_c + 1 \leq 2n_c+1 = 2(n - 1 + k)+1 = 2n - 1 + 2k = n_L + k$, (since $m_c \leq n_c$)

→ (ii) But to match $A_c(s)$ with the desired $A_{cd}(s)$, we only have to make sure $a_{c(nL-1)}, \ldots, a_{c0}$ equal to $a_{c(nK-1)}, \ldots, a_{c0}$, which represent n_L equations to solve

→ The number of free controller parameters $2n - 1+2k = n_L + k \geq$

The number of coefficients to match, which is n_L

→ As long as $k \geq 0$, infinite solution, when $k=0$, the solution is unique

Case 2: $n_L < 2n - 1$

→ Can not arbitrarily place the desired poles

Constrained Pole Placement

- **Controller with Integration**

Want $$D_c(s) = s\bar{D}_c(s)$$

Pole Placement Problem

$$D_{pn}(s)s\bar{D}_c(s) + N_{pn}(s)N_c(s) = A_{Cd}(s)$$

Equivalent Pole Placement Problem

$$\bar{D}_{pn}(s)\bar{D}_c(s) + N_{pn}(s)N_c(s) = A_{Cd}(s) \qquad \text{(PA2)}$$

which can be solved as before by assuming an equivalent fictitious plant of order n+1 and a new denominator of

$$\bar{D}_{pn}(s) = sD_{pn}(s)$$

Solution always exists if n_L is no less than $2n$. When $n_L=2n$, the solution is unique with order of $\bar{D}_c(s)$ being n-1 and order of $N_c(s)$ being n !

Constrained Pole Placement

- **Controller with Stable Pole/Zero Cancellation**

Q: Want to cancel some stable poles/zeros of plant TF

Example:

$$D_{pn}(s) = (s - p_c)\bar{D}_p(s) \qquad\qquad D_c(s) = (s - z_c)\bar{D}_c(s)$$
$$N_{pn}(s) = (s - z_c)\bar{N}_p(s) \quad \Rightarrow \quad N_c(s) = (s - p_c)\bar{N}_c(s)$$

Pole Placement Problem

$$= (s - p_c)(s - z_c)[\bar{D}_p(s)\bar{D}_c(s) + \bar{N}_p(s)\bar{N}_c(s)]$$

$$(s - p_c)\bar{D}_p(s)(s - z_c)\bar{D}_c(s) + (s - z_c)\bar{N}_p(s)(s - p_c)\bar{N}_c(s) = A_{Cd}(s)$$

which has a solution only if $A_{Cd}(s)$ contains the cancelled poles and zeros

$$A_{Cd}(s) = (s - p_c)(s - z_c)\bar{A}_{Cd}(s)$$

Equivalent Pole Placement Problem

$$\bar{D}_p(s)\bar{D}_c(s) + \bar{N}_p(s)\bar{N}_c(s) = \bar{A}_d^c(s) \qquad \text{(PA3)}$$

Cancelled poles/zeros remain as CL poles !

PID Controller via Pole Placement

- **Proper PID Controller Structure**

$$C(s) = K_P + \frac{K_I}{s} + \frac{K_D s}{\tau_D s + 1}$$

$$= K_P\left(1 + \frac{1}{T_r s} + \frac{T_d\, s}{\tau_D s + 1}\right)$$

$$= \frac{\left(K_D + K_P\tau_D\right)s^2 + \left(K_P + K_I\tau_D\right)s + K_I}{\tau_D s^2 + s}$$

$$T_r = \frac{K_P}{K_I}, \qquad \text{Reset time}$$

$$T_d = \frac{K_D}{K_P}, \qquad \text{Derivative time}$$

- **Equivalent Controller Form**

$$C(s) = \frac{b_{c2}s^2 + b_{c1}s + b_{c0}}{s^2 + a_{c1}s}$$

$$b_{c2} = \frac{K_D + K_p\tau_D}{\tau_D}, \quad b_{c0} = \frac{K_I}{\tau_D}$$

$$b_{c1} = \frac{K_p + K_I\tau_D}{\tau_D}, \quad a_{c1} = \frac{1}{\tau_D}$$

or

$$\tau_D = \frac{1}{a_{c1}}, \quad K_p = \left(b_{c1} - K_I\right)\tau_D$$

$$K_I = b_{c0}\tau_D, \quad K_D = \left(b_{c2} - K_p\right)\tau_D$$

PID Controller via Pole Placement

- **Determine PID Gains via Pole Placement**

$$D_{pn}(s)\left(s^2+a_{c1}s\right)+N_{pn}(s)\left(b_{c2}s^2+b_{c1}s+b_{c0}\right)=A_{Cd}(s)$$

Case 1: τ_D is pre-determined based on noise attenuation consideration

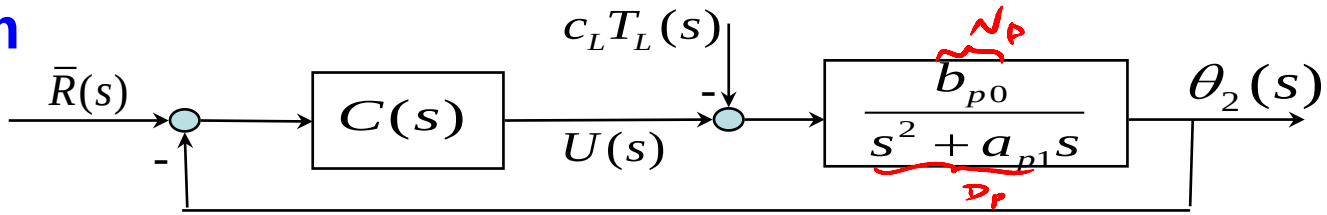
Only the Controller parameters b_{c2},b_{c1},b_{c0},or k_p,k_D and k_I are free to chosen. Thus, can only place three CL poles at the desired locations. So for plant with n>1,cannot arbitrarily place all CL poles.

Case 2: τ_D is automatically determined from pole placement design

Four controller parameters to choose, can place four CL poles at wish. So for plant with $n\leq 2$, we can arbitrarily place all CL poles.

PID Control of Rigid Belt ECP Emulator

- **System**



$b_{p0} = 623$
$a_{p1} = 3.5$

Would like to have magnitude of real parts of all CL poles larger than 20 for fast enough response if possible.

Case 1: τ_D is pre-determined

$$\frac{1}{\tau_D} \geq 10 \times 20 \Rightarrow \tau_D \leq \frac{1}{200} = 0.005,$$

$$\text{let } \tau_D = 0.005, \Rightarrow a_{c1} = 200$$

$$A_{CL}(s) = (s^2 + a_{p1}s)(s + a_{c1})s + b_{p0}(b_{c2}s^2 + b_{c1}s + b_{c0})$$

$$= s^4 + (a_{c1} + a_{p1})s^3 + (a_{p1}a_{c1} + b_{p0}b_{c2})s^2 + b_{p0}b_{c1}s + b_{p0}b_{c0}$$

As n_L=2+2=4**,** but we only have three controller parameters to choose, so let us choose

$$p_{1,2d} = -20 \pm 20j, p_{3d} = -40, p_{4,d} = -\alpha$$

PID Control of Rigid Belt ECP Emulator

$$
\begin{aligned}
A_{CLd}(s) &= (s^2 + a_{p1}s)(s + a_{c1})s + b_{p0}(b_{c2}s^2 + b_{c1}s + b_{c0}) \\
&= s^4 + (a_{c1} + a_{p1})s^3 + (a_{p1}a_{c1} + b_{p0}b_{c2})s^2 + b_{p0}b_{c1}s + b_{p0}b_{c0} \\
&= s^4 + 203.5s^3 + (700 + 623b_{c2})s^2 + 623b_{c1}s + 623b_{c0} \\
&= (s + 20 - 20j)(s + 20 + 20j)(s + 40)(s + \alpha)
\end{aligned}
$$

$$
\Rightarrow \begin{cases} \alpha = 123.5 \\ b_{c2} = 18.6 \\ b_{c1} = 527 \\ b_{c0} = 6340 \end{cases}
$$

PID Control of Rigid Belt ECP Emulator

Case 2: τ_D is automatically determined

Since we can arbitrarily place all CL poles, let us choose

$$p_{1,2d} = -20 \pm 20j,\ p_{3,4d} = -40$$

$$\begin{aligned}\Rightarrow A_{CLd}(s) &= (s-(-20+20j))(s-(-20-20j))(s-(-40))(s-(-40)) \\ &= [(s+20)^2+20^2](s+40)^2 \\ &= s^4+120s^3+5600s^2+128000s+1280000\end{aligned}$$

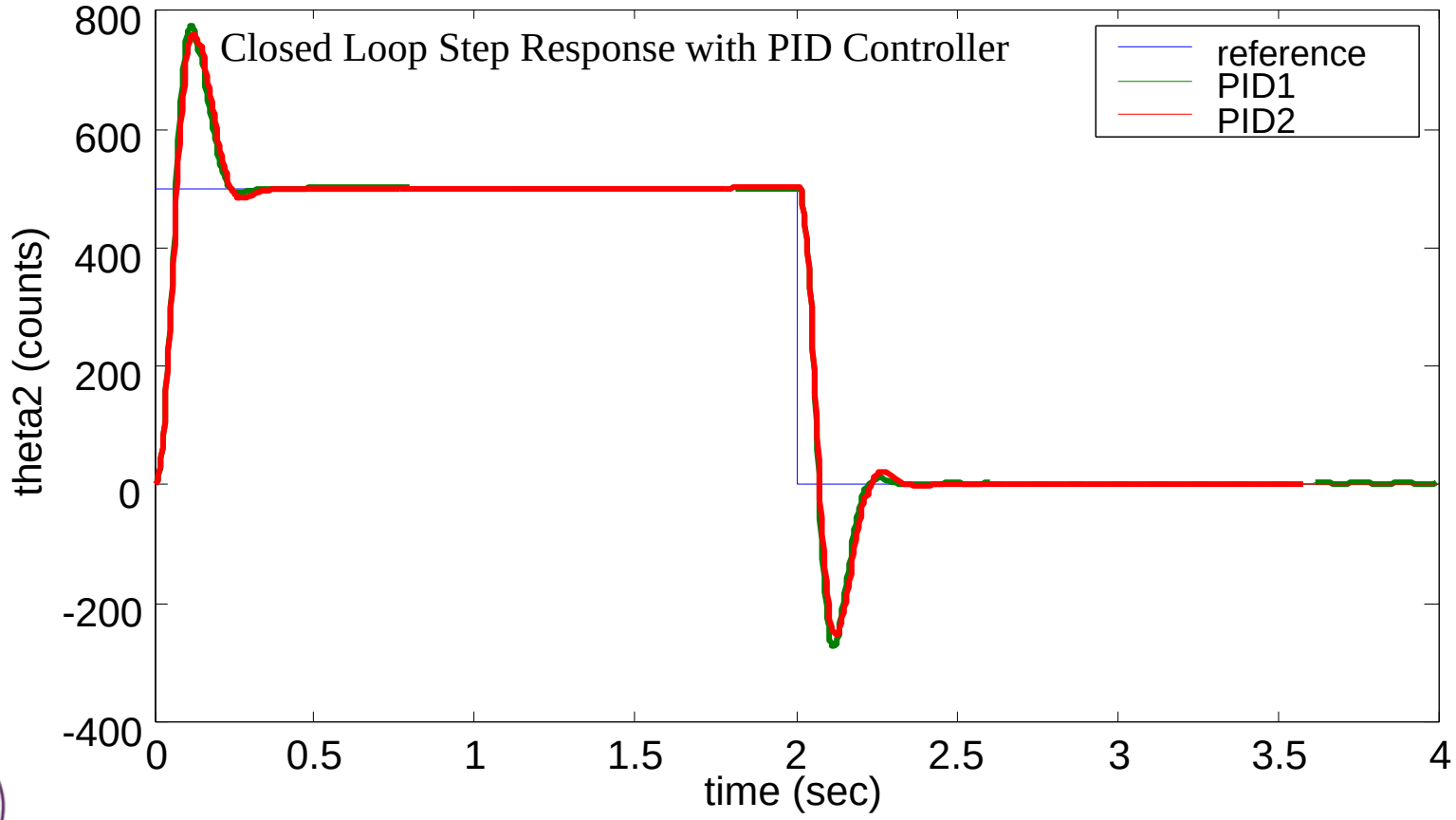
Actual CLCP：

$$A_{CL}(s) = \underbrace{(s^2+a_{p1}s)}_{D_p}\underbrace{(s^2+a_{c1}s)}_{D_c} + \underbrace{b_{p0}}_{N_p}(b_{c2}s^2+b_{c1}s+b_{c0})$$

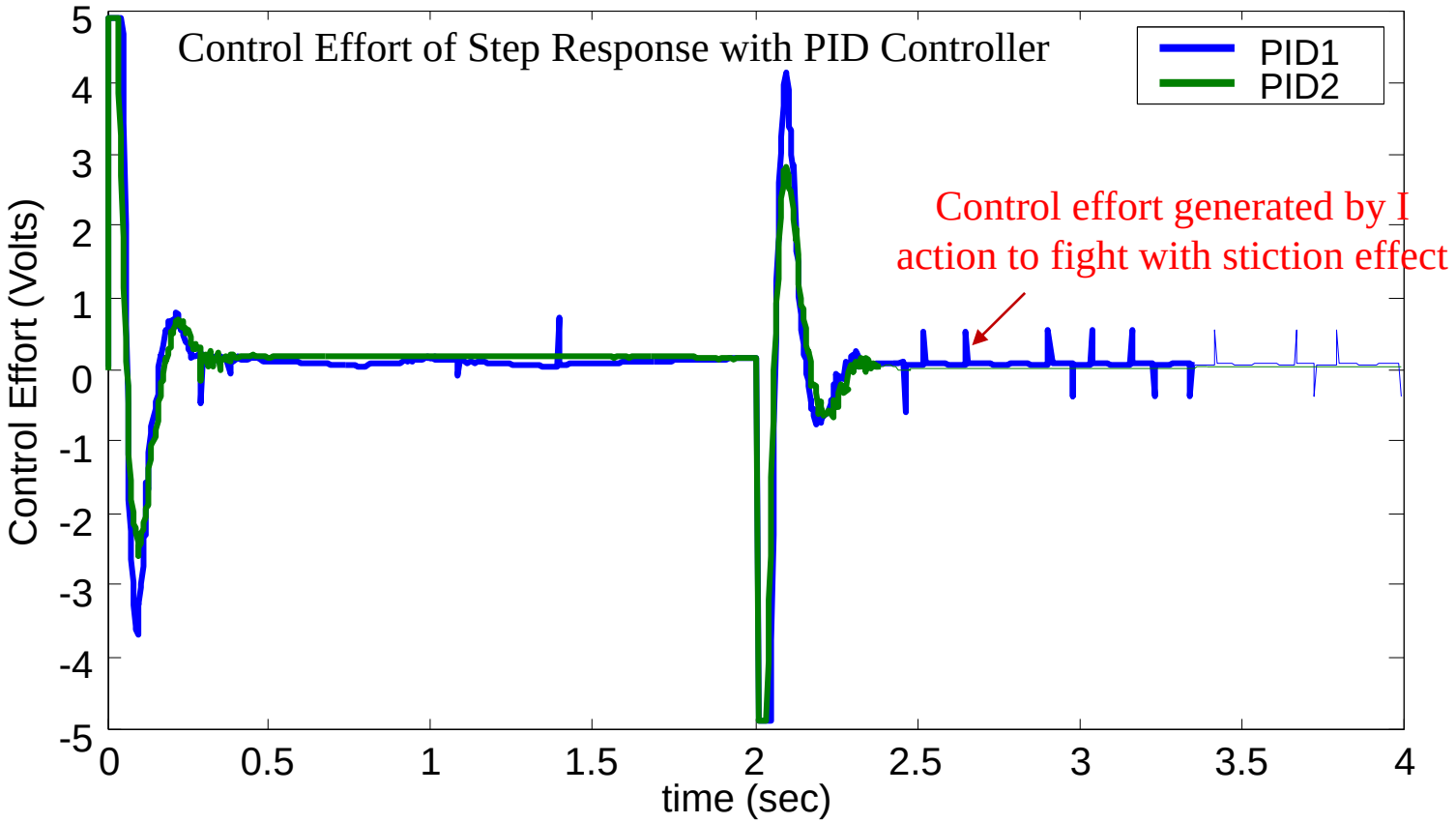
$$= s^4+(a_{p1}+a_{c1})s^3+(a_{p1}a_{c1}+b_{p0}b_{c2})s^2+b_{p0}b_{c1}s+b_{p0}b_{c0}$$

$$\Rightarrow \begin{cases} a_{p1}+a_{c1}=120 \\ a_{p1}a_{c1}+b_{p0}b_{c2}=5600 \\ b_{p0}b_{c1}=128000 \\ b_{p0}b_{c0}=1280000 \end{cases} \Rightarrow \begin{cases} a_{c1}=116.5 \\ b_{c2}=8.33 \\ b_{c1}=205 \\ b_{c0}=2054 \end{cases}$$

PID Control Experimental Results



PID Control Experimental Results



Main Contents of Course

Mechanical | Electrical | Electro-Mechanical | Fluid | Thermal

Dynamic Modeling

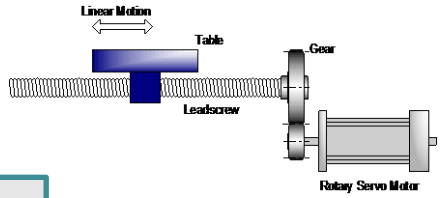
Structural selection

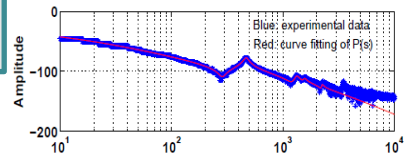
System Identification

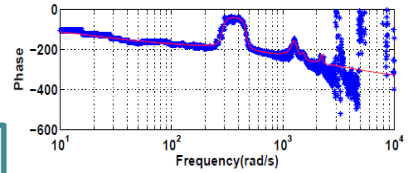
Parameter estimation

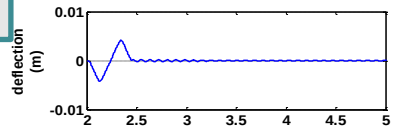
Control System Design

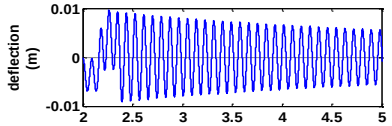
Root-locus design | Pole placement design | Frequency design | Architectural issues | Design limitations | State-feedback











Formulation of State Feedback Design

- **System (Plant)**

$$\dot{\underline{x}} = \underline{A}\,\underline{x} + \underline{B}u \quad \Rightarrow \quad P(s) = \frac{Y(s)}{U(s)} = \underline{C}\left(s\underline{I} - \underline{A}\right)^{-1}\underline{B} = \frac{1}{|sI - A|}C.Adj\{sI - A\}.B$$

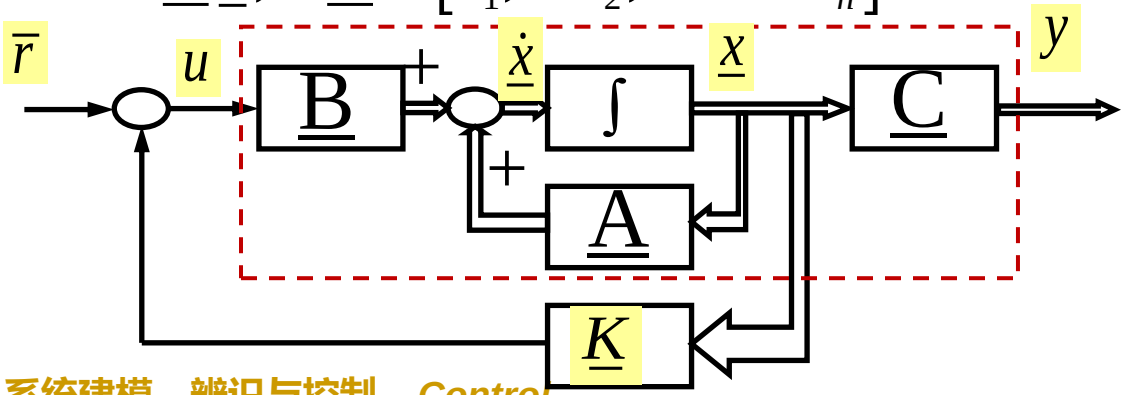
$$y = \underline{C}\,\underline{x}$$

Open-Loop Poles : $|sI - A| = 0$

Open-Loop Zeros : $C.Adj\{sI - A\}.B = 0$

- **Full State Feedback Control Law**

$$u = \bar{r} - \underline{K}\,\underline{x}, \quad \underline{K} = \left[k_1, \quad k_2, \quad \ldots \quad k_n\right]$$



Formulation of State Feedback Design

- **Closed-Loop System**

$$\dot{\underline{x}} = \left(\underline{A} - \underline{B}\underline{K}\right)\underline{x} + \underline{B}\,\bar{r}$$

$$y = \underline{C}\,\underline{x}$$

$$\Leftarrow \dot{x} = \underline{A}x + \underline{B}u = \underline{A}x + \underline{B}(-Kx + \bar{r}) = (\underline{A} - \underline{B}\,\underline{K})x + \underline{B}\bar{r}$$

- **Closed-Loop Transfer Function**

$$\frac{Y(s)}{\bar{R}(s)} = \underline{C}\left(s\underline{I} - \underline{A} + \underline{B}\underline{K}\right)^{-1}\underline{B} = \frac{\underline{C}\,Adj\left\{s\underline{I} - \underline{A} + \underline{B}\underline{K}\right\}\underline{B}}{\left|s\underline{I} - \underline{A} + \underline{B}\underline{K}\right|}$$

- **Closed-Loop Poles and Zeros**

CL Poles :

$$\left|s\underline{I} - \underline{A} + \underline{B}\underline{K}\right| = 0$$

CL Zeros :

$$\underline{C}\,Adj\left\{s\underline{I} - \underline{A} + \underline{B}\underline{K}\right\}\underline{B} = 0$$

Properties of State Feedback

- **State Feedback does not introduce additional dynamics in the loop but will change the locations of closed-loop poles**

Order of CL System :

$$n = \dim(A_{CL})$$

Order of OL System :

$$n = \dim(A_{OL})$$

- **State Feedback does not change zeros**

$$\underbrace{\underline{C}\,Adj\left\{s\underline{I} - \underline{A} + \underline{B}\,\underline{K}\right\}\underline{B}}_{=0,\quad CL\ zeros} = \underbrace{\underline{C}\,Adj\left\{s\underline{I} - \underline{A}\right\}\underline{B}}_{=0,\quad OL\ zeros}, \quad \forall \underline{K}$$

Open-loop zeros remain as closed-loop zeros !

Closed-Loop Pole Assignment Problem

- **Problem Formulation**

  Given the desired locations of closed-loop poles:

$$\phi(s) = (s - p_{CL1d})(s - p_{CL2d})\cdots(s - p_{CLnd})$$
$$= s^n + a_{CLn-1d}s^{n-1} + \cdots + a_{CL1d}s + a_{CL0d}$$

  Obtain a state feedback gain matrix (or vector) $\underline{K}$ such that the resulting closed-loop poles are at the above desired locations:

want

$$|s\underline{I} - \underline{A} + \underline{B}\underline{K}| = \phi(s)$$

Uncontrollable Systems

- **Controllable Decomposition:**

$$\dot{\underline{x}} = \underline{A}\,\underline{x} + \underline{B}\,u \qquad \underline{x} = \underline{T}\,\underline{z} \qquad \dot{\underline{z}} = \bar{\underline{A}}\,\underline{z} + \bar{\underline{B}}\,u$$

$$\underline{y} = \underline{C}\,\underline{x} + \underline{D}u \qquad \Rightarrow \qquad \underline{y} = \bar{\underline{C}}\,\underline{z} + \bar{\underline{D}}u$$

$$\bar{\underline{A}} = \underline{T}^{-1}\underline{A}\,\underline{T}$$
$$\bar{\underline{B}} = \underline{T}^{-1}\underline{B}$$
$$\bar{\underline{C}} = \underline{C}\,\underline{T}$$

$$\begin{bmatrix} \dot{\underline{z}}_c \\ \hdashline \dot{\underline{z}}_{uc} \end{bmatrix} = \left[\begin{array}{c:c} \bar{\underline{A}}_c & \bar{\underline{A}}_{cu} \\ \hdashline 0 & \bar{\underline{A}}_{uc} \end{array}\right] \begin{bmatrix} \underline{z}_c \\ \hdashline \underline{z}_{uc} \end{bmatrix} + \begin{bmatrix} \bar{\underline{B}}_c \\ \hdashline 0 \end{bmatrix} u$$

$\left[\bar{\underline{A}}_c, \quad \bar{\underline{B}}_c\right]$ is controllable

- **State Feedback Control Law:** $\bar{\underline{K}}_{c1} \in R^{1\times k}$ $\quad \bar{\underline{K}}_{c2} \in R^{1\times(n-k)}$

$$u = \bar{r} - \underline{K}\,\underline{x} = \bar{r} - \underline{K}\,\underline{T}\,\underline{z}, \qquad \bar{\underline{K}}_c = \underline{K}\,\underline{T} = \left[\bar{\underline{K}}_{c1} \quad \bar{\underline{K}}_{c2}\right]$$

$$\Rightarrow$$

$$\bar{\underline{A}} - \bar{\underline{B}}\bar{\underline{K}} = \left[\begin{array}{c:c} \bar{\underline{A}}_c & \bar{\underline{A}}_{cu} \\ \hdashline \underline{0} & \bar{\underline{A}}_{uc} \end{array}\right] - \begin{bmatrix} \bar{\underline{B}}_c \\ \hdashline \underline{0} \end{bmatrix} \left[\bar{\underline{K}}_{c1}, \quad \bar{\underline{K}}_{c2}\right] = \begin{bmatrix} \bar{\underline{A}}_c - \bar{\underline{B}}_c\bar{\underline{K}}_{c1} & \bar{\underline{A}}_{cu} - \bar{\underline{B}}_c\bar{\underline{K}}_{c2} \\ \underline{0} & \bar{\underline{A}}_{uc} \end{bmatrix}$$

Uncontrollable Systems

- **Closed-Loop Poles:**

$$\left|sI-\underline{\bar{A}}+\underline{\bar{B}}\,\underline{\bar{K}}\right|=\begin{vmatrix} sI-\underline{\bar{A}}_c+\underline{\bar{B}}_c\underline{\bar{K}}_{c1} & -\underline{\bar{A}}_{cu}+\underline{\bar{B}}_c\underline{\bar{K}}_{c2} \\ \underline{0} & sI-\underline{\bar{A}}_{uc} \end{vmatrix}$$

$$=\left|sI-\underline{\bar{A}}_c+\underline{\bar{B}}_c\underline{\bar{K}}_{c1}\right|\left|sI-\underline{\bar{A}}_{uc}\right|$$

Controllable CL Poles (freely assignable poles):

$\left|sI-(\underline{\bar{A}}_c-\underline{\bar{B}}_c\underline{\bar{K}}_{c1})\right|=0$ since $(\underline{\bar{A}}_c,\underline{\bar{B}}_c)$ is controllable,

$\Rightarrow k$ CL poles associated with controllable subspace can be arbitaryly assigned.

Uncontrollable CL Poles (unmovable poles):

$\left|sI-\underline{\bar{A}}_{uc}\right|=0$ can not be changed with any $\underline{\bar{K}}_c$,

$\Rightarrow n-k$ CL poles can not be changed.